Mathematics Division, School of Advanced Sciences and Languages
VIT Bhopal University
Winter Semester 2024 - 25
MAT2005-Transform Techniques and Difference Equations
Practice Set-3
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Laplace transform

1. Find the Laplace transforms of the given functions.

   (a) $f(t) = 6\mathbf{e}^{-5t} + \mathbf{e}^{3t} + 5t^3 - 9$
   Ans. $F(s) = \frac{6}{s+5} + \frac{1}{s-3} + \frac{30}{s^4} - \frac{9}{s}$

   (b) $g(t) = 4\cos(4t) - 9\sin(4t) + 2\cos(10t)$
   Ans. $G(s) = \frac{4s}{s^2+16} - \frac{36}{s^2+16} + \frac{2s}{s^2+100}$

   (c) $h(t) = 3\sinh(2t) + 3\sin(2t)$
   Ans. $H(s) = \frac{6}{s^2-4} + \frac{6}{s^2+4}$

   (d) $g(t) = \mathbf{e}^{3t} + \cos(6t) - \mathbf{e}^{3t}\cos(6t)$
   Ans. $\frac{1}{s-3} + \frac{s}{s^2+36} - \frac{s-3}{(s-3)^2+36}$

2. Find the transform of each of the following functions.

   (a) $f(t) = t\cosh(3t)$
   Ans. $F(s) = \frac{s^2+9}{(s^2-9)^2}$

   (b) $h(t) = t^2\sin(2t)$
   Ans. $H(s) = \frac{12s^2-16}{(s^2+4)^3}$

3. Find the Laplace transform of each function below.

   (a) $f(t) = t^3 - t^2 + 5t + 2$
   Ans. $F(s) = \dfrac{2s^3 + 5s^2 - 2s + 6}{s^4}$

   (b) $f(t) = -2\cos 6t + 5\sin 6t$
   Ans. $F(s) = \dfrac{-2s + 30}{s^2 + 36}$

   (c) $f(t) = 3e^{t} - 4e^{2t} + 2e^{-4t}$
   Ans. $F(s) = \dfrac{s^2 - 12s - 4}{(s-1)(s-2)(s+4)}$

   (d) $f(t) = 7t + 6e^{t} - 2e^{-t} - 10$
   Ans. $F(s) = \dfrac{-6s^3 + 15s^2 + 10s - 7}{s^4 - s^2}$

   (e) $f(t) = \sin 2t \sin 3t$
   Ans. $F(s) = \dfrac{s}{2(s^2+1)} - \dfrac{s}{2(s^2+25)}$

   (f) $f(t) = \sin 2t \cos 2t$
   Ans. $F(s) = \dfrac{2}{s^2+16}$

   (g) $f(t) = \cos^2 5t$
   Ans. $F(s) = \dfrac{1}{2}\left[\dfrac{1}{s} + \dfrac{s}{s^2+100}\right]$

   (h) $f(t) = t\cos^2 5t$
   Ans. $F(s) = \dfrac{1}{2}\left[\dfrac{1}{s^2} + \dfrac{s^2-100}{(s^2+100)^2}\right]$

   (i) $f(t) = te^{at}\cos bt$
   Ans. $F(s) = \dfrac{(s-a)^2 - b^2}{((s-a)^2+b^2)^2}$

   (j) $f(t) = te^{at}\sin bt$
   Ans. $F(s) = \dfrac{2b(s-a)}{((s-a)^2+b^2)^2}$

   (k) $f(t) = t^3\cos 2t$
   Ans. $F(s) = \dfrac{6s^4 - 144s^2 + 96}{(s^2+4)^4}$

   (l) $f(t) = \cos(\alpha t + \beta)$
   Ans. $F(s) = \dfrac{s\cos\beta - \alpha\sin\beta}{s^2+\alpha^2}$

   (m) $f(t) = \sin(\alpha t + \beta)$
   Ans. $F(s) = \dfrac{s\sin\beta + \alpha\cos\beta}{s^2+\alpha^2}$

   (n) $f(t) = f(t) = \sin^3 2t$
   Ans. $F(s) = \dfrac{48}{s^4 + 40s^2 + 144}$

Dr. Juhi Kesarwani and Dr. Ashish Kumar Kesarwany

Laplace transform of periodic functions

Find the Laplace transform of the following periodic function

1. $f(t) = t^2, \quad f(t+2) = f(t).$

   Ans. $\mathcal{L}\{f(t)\} = \dfrac{1}{1-e^{-2s}}\left[\dfrac{2}{s^3} - \left(\dfrac{4}{s} + \dfrac{4}{s^2} + \dfrac{2}{s^3}\right)e^{-2s}\right]$

2. $f(t) = \begin{cases} t & \text{if } 0 \le t < 1 \\ 2-t & \text{if } 1 \le t < 2 \end{cases} \qquad f(t+2) = f(t)$

   Ans. $\mathcal{L}\{f(t)\} = \dfrac{(1-e^{-s})^2}{s^2(1-e^{-2s})}$

3. $f(t) = \begin{cases} 1 & \text{if } 0 \le t < 1 \\ 0 & \text{if } 1 \le t < 2 \end{cases} \qquad f(t+2) = f(t)$

   Ans. $\mathcal{L}\{f(t)\} = \dfrac{1}{s(1+e^{-s})}$

Dr. Juhi Kesarwani

4. $f(t) = \begin{cases} t^2 & \text{if } 0 \le t < 2 \\ 2 & \text{if } 2 \le t < 3 \end{cases} \qquad f(t+3) = f(t)$

   Ans. $\mathcal{L}\{f(t)\} = \dfrac{1}{1+e^{-3s}}\left[\dfrac{2}{s^3} - e^{-2S}\left(\dfrac{2}{s^3} + \dfrac{4}{s^2} + \dfrac{2}{s}\right) + \dfrac{2}{s}e^{-3s}\right]$

and

5. $f(t) = \begin{cases} \dfrac{3t}{2} & \text{if } 0 \le t < 2 \\ 6 - \dfrac{3t}{2} & \text{if } 2 \le t < 4 \end{cases} \qquad f(t+4) = f(t)$

   Ans. $\mathcal{L}\{f(t)\} = \dfrac{3(1-e^{-2s})}{2s^2(1+e^{-2s})}$

Dr. Ashish Kumar Kesarwany

6. $f(t) = \begin{cases} 1 & \text{if } 0 \le t < 7 \\ -1 & \text{if } 7 \le t < 14 \end{cases} \qquad f(t+14) = f(t)$

   Ans. $\mathcal{L}\{f(t)\} = \dfrac{e^{7s}-1}{s(e^{7s}+1)}$

7. $f(t) = \begin{cases} 2 & \text{if } 0 \le t < 3 \\ -2 & \text{if } 3 \le t < 6 \end{cases} \qquad f(t+14) = f(t)$

   Ans. $\mathcal{L}\{f(t)\} = \dfrac{2e^{3s}-2}{s(e^{3s}+1)}$

8. $f(t) = 9t \qquad f(t+7) = f(t)$

   Ans. $\mathcal{L}\{f(t)\} = \dfrac{9e^{7s}-63s-9}{s^2(e^{7s}-1)}$

9. $f(t) = \sin \pi t$

   Ans. $\mathcal{L}\{f(t)\} = \dfrac{\pi}{s^2+\pi^2}$

Inverse Laplace Transform

1. Find the inverse Laplace transform of $F(s) = \dfrac{s-2}{s^2+4s+13}$.

   Ans. $f(t) = e^{-2t}\cos 3t - \frac{4}{3}e^{-2t}\sin 3t$

2. Find the inverse Laplace transform of $\dfrac{s^2}{(s+3)^3}$.

   Ans. $e^{-3t} - 6te^{-3t} + \frac{9}{2}t^2e^{-3t}$

3. Find the inverse transform of each of the following.

   (a) $F(s) = \dfrac{6}{s} - \dfrac{1}{s-8} + \dfrac{4}{s-3}$

   Ans. $f(t) = 6 - \mathbf{e}^{8t} + 4\mathbf{e}^{3t}$

   (b) $H(s) = \dfrac{19}{s+2} - \dfrac{1}{3s-5} + \dfrac{7}{s^5}$

   Ans. $h(t) = 19\mathbf{e}^{-2t} - \frac{1}{3}\mathbf{e}^{\frac{5t}{3}} + \frac{7}{24}t^4$

   (c) $F(s) = \dfrac{6s}{s^2+25} + \dfrac{3}{s^2+25}$

   Ans. $f(t) = 6\cos(5t) + \frac{3}{5}\sin(5t)$

   (d) $G(s) = \dfrac{8}{3s^2+12} + \dfrac{3}{s^2-49}$

   Ans. $g(t) = \frac{4}{3}\sin(2t) + \frac{3}{7}\sinh(7t)$

Dr. Juhi Kesarwani

4. Find the inverse transform of each of the following.

   (a) $F(s) = \dfrac{6s-5}{s^2+7}$

   Ans. $f(t) = 6\cos\left(\sqrt{7}t\right) - \frac{5}{\sqrt{7}}\sin\left(\sqrt{7}t\right)$

   (b) $F(s) = \dfrac{1-3s}{s^2+8s+21}$

   Ans. $f(t) = -3\mathbf{e}^{-4t}\cos\left(\sqrt{5}t\right) + \frac{13}{\sqrt{5}}\mathbf{e}^{-4t}\sin\left(\sqrt{5}t\right)$

and

   (c) $G(s) = \dfrac{3s-2}{2s^2-6s-2}$

   Ans. $g(t) = \frac{1}{2}\left(3\mathbf{e}^{\frac{3}{2}t}\cosh\left(\frac{\sqrt{13}}{2}t\right) + \frac{5}{\sqrt{13}}\mathbf{e}^{\frac{3}{2}t}\sinh\left(\frac{\sqrt{13}}{2}t\right)\right)$

   (d) $H(s) = \dfrac{s+7}{s^2-3s-10}$

   Ans. $h(t) = -\frac{5}{7}\mathbf{e}^{-2t} + \frac{12}{7}\mathbf{e}^{5t}$

Dr. Ashish Kumar Kesarwany

5. Find the inverse transform of each of the following.

   (a) $G(s) = \dfrac{86s-78}{(s+3)(s-4)(5s-1)}$

   Ans. $g(t) = -3\mathbf{e}^{-3t} + 2\mathbf{e}^{4t} + \mathbf{e}^{\frac{t}{5}}$

   (b) $F(s) = \dfrac{2-5s}{(s-6)(s^2+11)}$

   Ans. $f(t) = \frac{1}{47}\left(-28\mathbf{e}^{6t} + 28\cos\left(\sqrt{11}t\right) - \frac{67}{\sqrt{11}}\sin\left(\sqrt{11}t\right)\right)$

   (c) $G(s) = \dfrac{25}{s^3(s^2+4s+5)}$

   Ans. $g(t) = \frac{1}{5}\left(11 - 20t + \frac{25}{2}t^2 - 11\mathbf{e}^{-2t}\cos(t) - 2\mathbf{e}^{-2t}\sin(t)\right)$

6. Find the inverse Laplace transform of each function below.

   (a) $F(s) = \dfrac{4s+2}{s^2+6s+34}$

   Ans. $f(t) = 4e^{-3t}\cos 5t - 2e^{-3t}\sin 5t$

   (b) $F(s) = \dfrac{1}{(s-2)(s-4)(s-8)}$

   Ans. $f(t) = \dfrac{1}{12}e^{2t} - \dfrac{1}{8}e^{4t} + \dfrac{1}{24}e^{8t}$

(c) $F(s) = \dfrac{3s-7}{4s^2+1}$

Ans. $f(t) = \dfrac{3}{4}\cos\dfrac{t}{2} - \dfrac{7}{2}\sin\dfrac{t}{2}$

(d) $F(s) = \dfrac{s^2+s-6}{s^3+2s^2+s}$

Ans. $f(t) = 7e^{-t} + 6te^{-t} - 6$

(e) $F(s) = \dfrac{1}{s^4-81}$

Ans. $f(t) = \dfrac{1}{108}e^{3t} - \dfrac{1}{108}e^{-3t} - \dfrac{1}{54}\sin 3t$

(f) $F(s) = \dfrac{s^3}{s^4-16}$

Ans. $f(t) = \dfrac{1}{4}e^{2t} + \dfrac{1}{4}e^{-2t} - \dfrac{1}{2}\cos 2t$

(g) $F(s) = \dfrac{10}{s^4-s^3}$

Ans. $f(t) = 10e^t - 5t_1^20t - 10$

(h) $F(s) = \dfrac{12}{s^3-8}$

Ans. $f(t) = e^{2t} - e^{-t}\cos(\sqrt{3}t) - \sqrt{3}e^{-t}\sin(\sqrt{3}t)$

(i) $F(s) = \dfrac{1}{(s-\alpha)(s-\beta)}$

Ans. $f(t) = \dfrac{1}{\alpha-\beta}(e^{\alpha t} - e^{\beta t})$

7. Find the inverse Laplace transform of each of the following.

Dr. Juhi Kesarwani

(a) $H(s) = \dfrac{s\mathbf{e}^{-4s}}{(3s+2)(s-2)}$

Ans. $h(t) = u_4(t)f(t-4) = \begin{cases} f(t-4) & t \geq 4 \\ 0 & t < 4 \end{cases}$, **where** $f(t) = \frac{1}{12}\mathbf{e}^{-\frac{2t}{3}} + \frac{1}{4}\mathbf{e}^{2t}$

(b) $G(s) = \dfrac{5\mathbf{e}^{-6s} - 3\mathbf{e}^{-11s}}{(s+2)(s^2+9)}$

Ans. $g(t) = 5u_6(t)f(t-6) - 3u_{11}(t)f(t-11)$, **where** $f(t) = \frac{1}{13}\left(\mathbf{e}^{-2t} - \cos(3t) + \frac{2}{3}\sin(3t)\right)$

(c) $F(s) = \dfrac{4s+\mathbf{e}^{-s}}{(s-1)(s+2)}$

Ans. $f(t) = \frac{4}{3}\mathbf{e}^t + \frac{8}{3}\mathbf{e}^{-2t} + u_1(t)h(t-1)$, **where** $h(t) = \frac{1}{3}\mathbf{e}^t - \frac{1}{3}\mathbf{e}^{-2t}$

and

(d) $G(s) = \dfrac{3s + 8\mathbf{e}^{-20s} - 2s\mathbf{e}^{-3s} + 6\mathbf{e}^{-7s}}{s^2(s+3)}$

Ans. $g(t) = 3f(t) - 2u_3(t)f(t-3) + 8u_{20}(t)h(t-20) + 6u_7(t)h(t-7)$ **where** $f(t) = \frac{1}{3} - \frac{1}{3}\mathbf{e}^{-3t}$, **and** $h(t) = -\frac{1}{9} + \frac{1}{3}t + \frac{1}{9}\mathbf{e}^{-3t}$

Convolution theorem

Dr. Ashish Kumar Kesarwany

1. Show that $\int_0^t f(t-\tau)g(\tau)\,d\tau = \int_0^t f(\tau)g(t-\tau)\,d\tau$ by introducing the new variable of integration $x = t-\tau$ in the first integral.

2. Find $\mathcal{L}^{-1}\left\{\frac{s}{(s^2+1)(s-1)}\right\}$.

Ans. $\frac{2}{5}e^{2t} + \frac{1}{5}(\sin t - 2\cos t)$

3. Use the Convolution theorem to find the inverse Laplace transform of the following

(a) $F(s) = \dfrac{1}{s^2(s^2+4)}$

Hint. $f(t) = \dfrac{1}{2}\displaystyle\int_0^t \tau\sin 2(t-\tau)d\tau$

(b) $F(s) = \dfrac{s}{(s+2)(s^2+9)}$

Hint. $f(t) = \displaystyle\int_0^t e^{-2\tau}\cos 3(t-\tau)d\tau$

(c) $F(s) = \dfrac{s}{(s^2+4)(s^2+9)}$

Hint. $f(t) = \dfrac{1}{2}\displaystyle\int_0^t \sin 2\tau\cos 3(t-\tau)d\tau$

(d) $F(s) = \dfrac{s}{(s^2+1)^2}$

Hint. $f(t)\displaystyle\int_0^t \cos\tau\sin(t-\tau)d\tau$

(e) $F(s) = \dfrac{1}{s(s-a)}$

Hint. $f(t) = \displaystyle\int_0^t e^{a\tau} d\tau$

(f) $F(s) = \dfrac{1}{(s+1)(s^2+2s+2)}$

Hint. $f(t) = e^{-t} \displaystyle\int_0^t \sin(t-\tau) d\tau$

(g) $F(s) = \dfrac{1}{(s+1)^2(s^2+4s+5)}$

Hint. $f(t) = e^{-2t} \displaystyle\int_0^t \tau e^{\tau} \sin(t-\tau) d\tau$

(h) $F(s) = \dfrac{1}{(s-1)^3(s+2)^2}$

Hint. $f(t) = \dfrac{e^{-2t}}{2} \displaystyle\int_0^t \tau^2 (t-\tau) e^{3\tau} d\tau$

(i) $F(s) = \dfrac{s-1}{s^2(s^2-2s+2)}$

Hint. $f(t) = \displaystyle\int_0^t (t-\tau) e^{\tau} \cos\tau d\tau$

(j) $F(s) = \dfrac{s(s+3)}{(s^2+4)(s^2+6s+10)}$

Hint. $f(t) = \displaystyle\int_0^t e^{-3\tau} \cos\tau \cos 2(t-\tau) d\tau$

(k) $F(s) = \dfrac{1}{(s-3)^5 s^6}$

Hint. $f(t) = \dfrac{1}{4!5!} \displaystyle\int_0^t \tau^4 (t-\tau)^5 e^{3\tau} d\tau$

(l) $F(s) = \dfrac{1}{(s-1)^3(s^2+4)}$

Hint. $f(t) = \dfrac{1}{4} \displaystyle\int_0^t \tau^2 e^{\tau} \sin 2(t-\tau) d\tau$

(m) $F(s) = \dfrac{1}{s^2(s-2)^3}$

Hint. $f(t) = \dfrac{1}{2} \displaystyle\int_0^t \tau (t-\tau)^2 e^{2(t-\tau)} d\tau$

(n) $F(s) = \dfrac{1}{s^7(s-2)^6}$

Hint. $f(t) = \dfrac{1}{5!6!} \displaystyle\int_0^t (t-\tau)^5 e^{2(t-\tau)} \tau^6 d\tau$

4. Use the convolution theorem to find the Laplace transform of the following

(a) $f(t) = \displaystyle\int_0^t \sin a\tau \cos b(t-\tau)\, d\tau$

Ans. $F(s) = \dfrac{as}{(s^2+a^2)(s^2+b^2)}$

(b) $f(t) = \displaystyle\int_0^t e^{\tau} \sin a(t-\tau)\, d\tau$

Ans. $F(s) = \dfrac{as}{(s^2-a^2)^2}$

(c) $f(t) = \displaystyle\int_0^t \sinh a\tau \cosh a(t-\tau)\, d\tau$

Ans. $F(s) = \dfrac{as}{(s^2-a^2)^2}$

(d) $f(t) = \displaystyle\int_0^t \tau(t-\tau) \sin\omega\tau \cos\omega(t-\tau)\, d\tau$

Ans. $F(s) = \dfrac{2\omega s(s^2-\omega^2)}{(s^2+\omega^2)^4}$

(e) $f(t) = e^t \displaystyle\int_0^t \sin\omega\tau \cos\omega(t-\tau)\, d\tau$

Ans. $F(s) = \dfrac{(s-1)\omega}{((s-1)^2+\omega^2)^2}$

(f) $f(t) = e^t \displaystyle\int_0^t \tau^2 (t-\tau) e^{\tau}\, d\tau$

Ans. $F(s) = \dfrac{2}{(s-2)^3(s-1)^2}$

Dr. Juhi Kesarwani and Dr. Ashish Kumar Kesarwany

Solving ODEs using Laplace transform

1. Find the solution of the following first order ODE

$$\frac{dy}{dt} + 3y = 0, \quad y(0) = 1$$

Ans. $y(t) = e^{-2t}$

2. Solve the following IVP.

$$y'' - 6y' + 15y = 2\sin(3t), \qquad y(0) = -1 \quad y'(0) = -4$$

Ans. $y(t) = \frac{1}{10}\left(\cos(3t) + \frac{1}{3}\sin(3t) - 11e^{3t}\cos\left(\sqrt{6}t\right) - \frac{8}{\sqrt{6}}e^{3t}\sin\left(\sqrt{6}t\right)\right)$

3. Solve the following IVP.

$$y'' - 10y' + 9y = 5t, \qquad y(0) = -1 \quad y'(0) = 2$$

Ans. $y(t) = \frac{50}{81} + \frac{5}{9}t + \frac{31}{81}e^{9t} - 2e^{t}$

4. Find the solution of the following second order ODE

$$\frac{d^2y}{dt^2} + y = t, \quad y(0) = 1, \quad y'(0) = -2$$

Ans. $y(t) = t + \cos t - 3\sin t$

5. Use the method of Laplace transforms to solve each of the following IVPs.

(a) $y' + 10y = t^2, \qquad y(0) = 0$

Ans. $y = \frac{t^2}{100} - \frac{t}{50} + \frac{1}{500} - \frac{1}{500}e^{-10t}$

(b) $y' + 2y = t^{-t}, \qquad y(0) = 2$

Ans. $y = te^{-t} - e^{-t} + 3e^{-2t}$

(c) $y' - 6y = 2\sin 3t, \qquad y(0) = -1$

Ans. $y = \frac{-13}{15}e^{6t} - \frac{2}{15}\cos 3t - \frac{4}{15}\sin 3t$

(d) $y'' + 2y' = te^{-t}, \qquad y(0) = 6 \qquad y'(0) = -1$

Ans. $y = -te^{-t} + 6$

(e) $y'' + 4y' - 5y = 0,, \qquad y(0) = 5 \qquad y'(0) = -1$

Ans. $y = 4e^{t} + e^{-5t}$

(f) $y'' - 2y' + y = 2t - 3, \qquad y(0) = 5 \qquad y'(0) = 11$

Ans. $y = 4e^{t} + 5te^{t} + 2t + 1$

(g) $y'' + 8y' + 25y = 13e^{-2t}, \qquad y(0) = -1 \qquad y'(0) = 18$

Ans. $y = -2e^{-4t}\cos 3t + 4e^{-4t}\sin 3t + e^{-2t}$

(h) $y'' + 6y' + 34y = 0, \qquad y(0) = -1 \qquad y'(0) = 13$

Ans. $y = -e^{-3t}\cos 5t + 2e^{-3t}\sin 5t$

(i) $y'' + 4y = 8\cos 2t - 8e^{-2t}, \qquad y(0) = -2 \qquad y'(0) = 0$

Ans. $y = -\cos 2t - \sin 2t + 2t\sin 2t - e^{-2t}$

(j) $y'' - 4y' + 4y = 8t^2 - 16t + 4, \qquad y(0) = 0 \qquad y'(0) = 1$

Ans. $y = te^{2t} + 2t^2$

(k) $y'' - 4y' - 5y = 3t^3, \qquad y(0)3 = \qquad y'(0)3 =$

Ans. $y = e^{5t} + 2e^{-t} + 3t^3$

(l) $y'' + 3y'' + 3y' + y = 0,, \qquad y(0) = 7 \qquad y'(0) = -7 \qquad y''(0) = 11$

Ans. $y = 7e^{-t} + 2t^2e^{-t}$

(m) $y''' + 4y'' - 5y' = 0,, \qquad y(0) = 4 \qquad y'(0) = -7 \qquad y''(0) = 23$

Ans. $y = 5 - 2e^{t} + e^{-5t}$

(n) $y''' - y'' + 4y' - 4y = 26e^{3t}, \qquad y(0) = -2 \qquad y'(0) = 3 \qquad y''(0) = 1$

Ans. $y = -4e^{t} + e^{3t} + \cos 2t + 2\sin 2t$

Dr. Juhi Kesarwani and Dr. Ashish Kumar Kesarwany

Miscellaneous Problems

1. Solve the following BVP

$$y'' + 4y' = \cos(t-3) + 4t, \qquad y(3) = 0 \quad y'(3) = 7$$

Ans. $y(t) = \frac{1}{2}t^2 - \frac{1}{4}t - \frac{43}{16} - \frac{273}{272}e^{-4(t-3)} + \frac{1}{17}(4\sin(t-3) - \cos(t-3))$

2. Express the following function in terms of unit step function

$$f(t) = \begin{cases} (t-1) & \text{if } 1 \le t < 2 \\ (3-t) & \text{if } 2 \le t < 3 \end{cases}$$

Ans. $f(t) = (t-1)u_1(t) - 2(t-2)u_2(t) + (t-3)u_3(t)$ **and** $\mathcal{L}\{f(t)\} = \frac{e^{-s}}{s^2} - 2\frac{e^{-2s}}{s^2} + \frac{e^{-3s}}{s^2}$

3. Find the Laplace transforms of each of the following functions

(a) $f(t) = \dfrac{\sin t}{t}$

Ans. $\frac{\pi}{2} - \tan^{-1} s$

(b) $g(t) = \dfrac{1-\cos 2t}{t}$

Ans. $\ln \dfrac{\sqrt{s^2+4}}{s}$

(c) $h(t) = \dfrac{e^{4t} - e^{-3t}}{t}$

Ans. $\ln \dfrac{s+3}{s-4}$

(d) $f(t) = \dfrac{\sin^2 t}{t}$

Ans. $\frac{1}{4}\ln\left(\dfrac{s^2+4}{s^2}\right)$

(e) $g(t) = \dfrac{\cos 4t - \cos 5t}{t}$

Ans. $\frac{1}{2}\ln\dfrac{s^2+25}{s^2+16}$

(f) $h(t) = \dfrac{1-e^t}{t}$

Ans. $\ln \dfrac{s-1}{s}$

(g) $f(t) = \dfrac{e^{-t}\sin t}{t}$

Ans . $\frac{\pi}{2} - \tan^{-1}(s+1)$

4. Find the Laplace transform of each of the following function

(a) $f(t) = \begin{cases} \sin(t-1) & t \ge 1 \\ 0 & t < 0 \end{cases}$

Ans. $\dfrac{e^{-s}}{s^2+1}$

(b) $g(t) = \begin{cases} (t-1)\sin(t-1) & t \ge 1 \\ 0 & t < 0 \end{cases}$

Ans. $\dfrac{2se^{-s}}{(s^2+1)^2}$

(c) $h(t) = \sin t\ u_\pi(t)$

Ans. $-\dfrac{e^{-\pi s}}{s^2+1}$

(d) $f(t) = (t-1)^2 u_1(t)$

Ans. $\dfrac{2e^{-s}}{s^2}$

5. Find the inverse Laplace transform of the following functions.

(a) $\ln\left(\dfrac{s-a}{s-b}\right)$

Ans. $\dfrac{e^{bt} - e^{at}}{t}$

(b) $\ln\left(\dfrac{s^2+a^2}{\sqrt{s+b}}\right)$.

Ans. $\dfrac{e^{-bt}}{2t} - \dfrac{2\cos at}{t}$

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